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distance from the ground to the center of the wheel's hub is .28 m. The damper for the car has been damaged and only gives us 265 N·s/m. We'll determine what the maximum vibration amplitude will be for this system and the vehicle speed at which it will occur.

Solution Using (2.9.6), we write

$$\bar{x} = \frac{(.002)(15)\omega^2}{-(35)\omega^2 + 200,000 + 265\omega i} \tag{2.9.13}$$

Note that the mass in the numerator is 15 kg because this is the total rotating mass, whereas the mass in the denominator is 35 kg (the total mass, including rotating and nonrotating parts). To find the maximal response, we can differentiate $\bar{x}$ with respect to ω, set the resultant to zero, and solve for ω. An easier way to get the same result is to square $\bar{x}$ and differentiate with respect to ω^2. The peak response will occur at the same frequency, but the mathematical manipulations are simpler in the second approach.

Carrying through will ultimately yield

$$\omega_{rp} = \frac{\omega_n}{\sqrt{1 - 2\zeta^2}}$$

where the subscript rp stands for rotating peak.

For this problem $\omega_n^2 = \frac{200,000}{35}$ and so $\omega_n = 75.59$ rad/s. From $\frac{c}{m} = 2\zeta\omega_n$, we can find that $\zeta = .05$. Thus $\omega_{rp} = 75.78$ rad/s. Using this gives us a peak response magnitude of

$$|\bar{x}| = 8.57 \text{ mm}$$

If we can now find the corresponding velocity of the car, we're done. Since the distance from the ground to the hub of the wheel is .28 m, we can find the car's velocity from

$$v = \omega r = (75.78 \text{ rad/s})(.28 \text{ m}) = 21.2 \text{ m/s}$$

or 47 mph, for those of you who prefer English units.

2.10 IDENTIFICATION OF DAMPING AND NATURAL FREQUENCY

A topic of great practical importance is that of identification. In the real world, nobody is going to tell you the damping factor or natural frequency for the system you're examining. Thus, it's up to you to figure these quantities out for yourself. The field of system identification is a huge one and can easily fill a couple of courses. Although we can't go over all this material in this book, we'll look at a few useful techniques that can give us good approximations to the quantities we're looking for.

The first response type we'll look at is the unforced system response. If the damping is relatively low, we can get a good estimate of both the natural frequency and the damping factor just by examining how the response decays after an initial excitation. If the damping is zero, we know that the natural frequency is simply the frequency of oscillation of the system. And if the damping is small, then the damped oscillation frequency ω_d ($\omega_d = \omega_n\sqrt{1 - \zeta^2}$) is very close to ω_n. We can, however, do much better than simply approximating our answers. Let's recall our solution to the unforced problem (1.4.15). If we include only the cosine term (implying an initial displacement with zero

initial velocity), then at $t = 0$ the response is equal to b_1. Next, look at the response one period later, i.e., at $t = t_p = \frac{2\pi}{\omega_d}$. Now the response is

$$x(t_p) = b_1 e^{\frac{-2\pi\zeta}{\sqrt{1-\zeta^2}}} \tag{2.10.1}$$

If we take the ratio of these two values, we obtain

$$\frac{x(0)}{x(t_p)} = e^{\frac{2\pi\zeta}{\sqrt{1-\zeta^2}}} \tag{2.10.2}$$

Taking the natural log of both sides yields

$$\ln\left(\frac{x(0)}{x(t_p)}\right) = \frac{2\pi\zeta}{\sqrt{1-\zeta^2}} \tag{2.10.3}$$

This result is known as the logarithmic decrement and is usually denoted by σ,

$$\sigma \equiv \ln\left(\frac{x(0)}{x(t_p)}\right) \tag{2.10.4}$$

Although we derived this from a signal that was maximal at $t = 0$, it can easily be found by taking the ratio of any two successive peaks. Once we have σ, it becomes a simple matter to solve for ζ,

$$\zeta = \frac{\sigma}{\sqrt{4\pi^2 + \sigma^2}} \tag{2.10.5}$$

This expression can be further simplified if the damping factor is small. In this case the denominator is closely approximated by 2π and our expression for ζ becomes

$$\zeta = \frac{\sigma}{2\pi} \tag{2.10.6}$$

If we wish to use more oscillations, the formula need be changed only slightly. As long as we're letting an integer number of oscillations elapse between samples, the cosine term is always equal to 1.0. Thus, the only change is that the response will decay due to the exponential envelope. If we wait for n periods, the response amplitude will be

$$x(nt_p) = b_1 e^{\frac{-2n\pi\zeta}{\sqrt{1-\zeta^2}}} \tag{2.10.7}$$

In this case, the ratio becomes

$$\frac{x(0)}{x(nt_p)} = e^{\frac{2n\pi\zeta}{\sqrt{1-\zeta^2}}} \tag{2.10.8}$$

Taking the natural log of this ratio leads to

$$\ln\left(\frac{x(0)}{x(nt_p)}\right) = \frac{2n\pi\zeta}{\sqrt{1-\zeta^2}} \tag{2.10.9}$$

This is equal to n times the log decrement as determined from two successive peaks. Thus

$$\sigma = \frac{1}{n} \ln\left(\frac{x(0)}{x(nt_p)}\right) \tag{2.10.10}$$

Once σ has been determined, we can find ζ using either (2.10.5) or (2.10.6). This way of determining ζ is also a very good means of determining whether the damping for your particular system is linear. If it *is* linear, then the damping factor found between peaks 1 and 2 of the response should be the same as that found between peaks 2 and 3. If the results differ, this implies that the damping isn't linear.

Example 2.13

Problem Consider the response shown in Figure 2.26. This response corresponds to a spring-mass-damper that's been struck by an impact hammer, resulting in the vibrations shown. Determine the damping factor by means of the log decrement.

Solution The first peak looks to be about .093 and the second is around .068. Thus $\sigma = \ln\left(\frac{.093}{.068}\right) = .313$. Equation (2.10.5) lets us solve for ζ and leads to an estimate of $\zeta = .05$.

If we try and estimate ζ from two successive peaks, we can use the first peak of .093 and the third peak of .05. This will produce a σ of .310, which will give a ζ of .049.

Both these estimates are excellent ones: the equation of motion used to generate the illustrated response was

$$\ddot{x} + \dot{x} + 100x = 0$$

for which $\omega_n = 10$ rad/s and $\zeta = .05$.

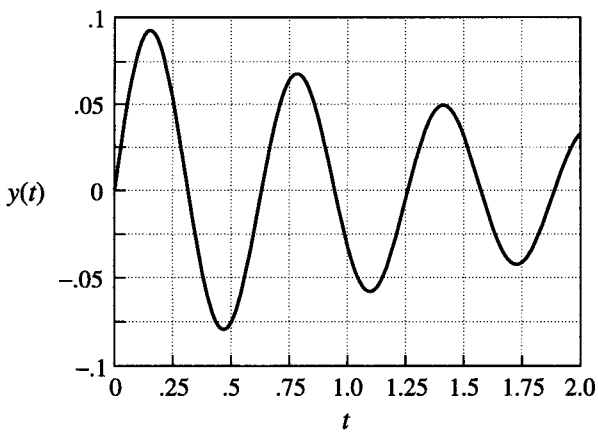


Figure 2.26 Free response of SDOF system

Example 2.14

Problem You've been given the plot shown in Figure 2.27, which displays the displacement response of a vibrating microstructure. Estimate the damping factor from at least two data pairs and comment on whether the system damping is or is not linear.

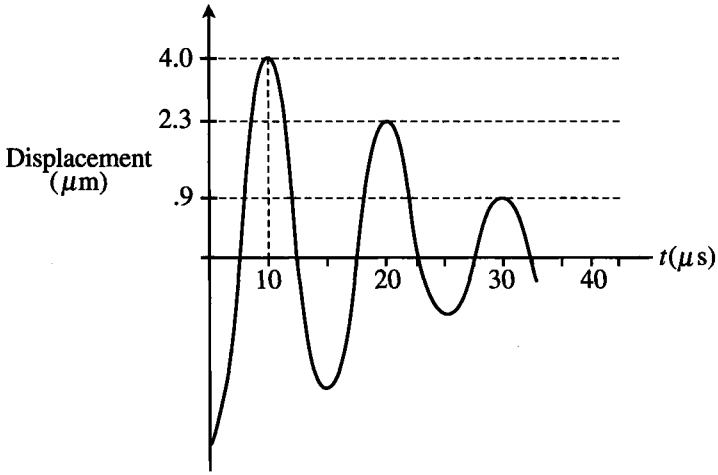


Figure 2.27 Displacement response of a vibrating microstructure

Solution Using the first two peaks we have

$$\sigma = \ln\left(\frac{4}{2.3}\right) = .5534 \quad \text{from (2.10.4)}$$

and

$$\zeta_{1-2} = \frac{.5534}{\sqrt{4\pi^2 + (.5534)^2}} = .0877 \quad \text{from (2.10.5)}$$

Using the second and third peaks we'll obtain

$$\sigma = \ln\left(\frac{2.3}{.9}\right) = .9383$$

and

$$\zeta_{2-3} = \frac{.9383}{\sqrt{4\pi^2 + (.9383)^2}} = .1477$$

ζ_{1-2} and ζ_{2-3} differ appreciably, indicating that the actual damping mechanism is nonlinear.

What we've now seen is that a free vibration test will allow us to identify the damping factor in a single-degree-of-freedom system by means of the log decrement. Once the damping factor is known, we can find the natural frequency from a knowledge of the damped frequency of oscillation ($\omega_d = \omega_n\sqrt{1-\zeta^2}$). We're not limited, however, to free vibration tests. We can glean the same information from a forced test. In these tests, we're looking for the transfer functions of the system. We can obtain these transfer functions in a number of ways and we'll look at some of them in Chapter 8. For now we'll just assume that somehow we've obtained them and they're available for use in our identification tests.

If you look at Figure 2.28*a* you'll note that, for the direct force excited system, the phase angle goes through –90 degrees exactly at the system's natural frequency, regardless of the damping value.

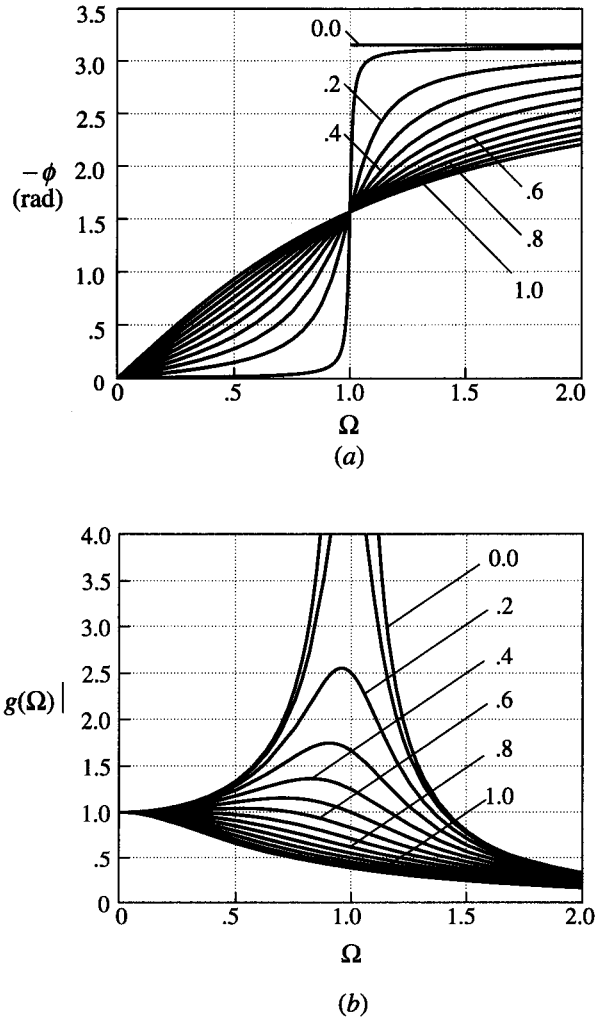


Figure 2.28 (*a*) Phase and (*b*) amplitude response of a direct force excited, spring-mass-damper system (varying ζ values indicated by numbers on plot)

Thus, all we need do is look at the phase, determine when it goes through -90 degrees, and read off the frequency at which this occurs. This will give us ω_n.

Interestingly, there are even more indicators of ω_n and ζ. The first one is found by comparing the actual peak amplitude response of the system to the response at zero frequency. From (2.6.29), repeated here, we know that the transfer function of a direct force excited system is given by

$$|g(\omega)| = \frac{1}{m\sqrt{(\omega_n^2 - \omega^2)^2 + (2\zeta\omega_n\omega)^2}} \tag{2.10.11}$$

Thinking back to your first calculus class, you'll recall that the slope of a curve is zero at a local maximum. Thus, to determine the frequency at which our amplitude response curve is a maximum

(which we'll call ω_p), we need only differentiate with respect to ω and set the result equal to zero. Actually, although this approach will give us the correct answer, we can simplify matters even further. Rather than differentiating with respect to ω, we can differentiate with respect to ω^2, since only even powers of ω show up in the equation. This will give us ω_p^2 instead of ω_p. We can easily take the square root to find ω_p. However we can simplify yet further. Taking the derivative of the reciprocal of the square root of a function is somewhat complicated. So instead, let's square $g(\omega)$. Since the function is positive everywhere anyway, this won't change our results. And finally, rather than looking at $g^2(\omega)$, let's differentiate $\frac{1}{g^2(\omega)}$. Thus, instead of looking for the maximum of $g(\omega)$ (shown in Figure 2.29), we'll be looking for the minimum of $h(\omega^2) \equiv \frac{1}{g^2(\omega)}$ (Figure 2.30). Thus we want

$$\left.\frac{dh(\omega^2)}{d\omega^2}\right|_{\omega=\omega_p} = 0$$

or

$$2(\omega_n^2 - \omega_p^2)(-1) + (2\zeta\omega_n)^2 = 0$$

Solving for ω_p, we obtain

$$\omega_p = \omega_n\sqrt{1 - 2\zeta^2} \tag{2.10.12}$$

For small values of damping, ω_p is approximately equal to ω_n. So another way of estimating ω_n for lightly damped systems is to use the frequency at which the peak response occurs.

We can also use this result to verify what we can see from Figure 2.28*b*. We've already mentioned that, as ω increases from zero and approaches ω_n, the amplitude response increases if the damping is small but decreases if the damping is large. We can also see from Figure 2.26*b* that the response peak moves to the left as the damping increases.

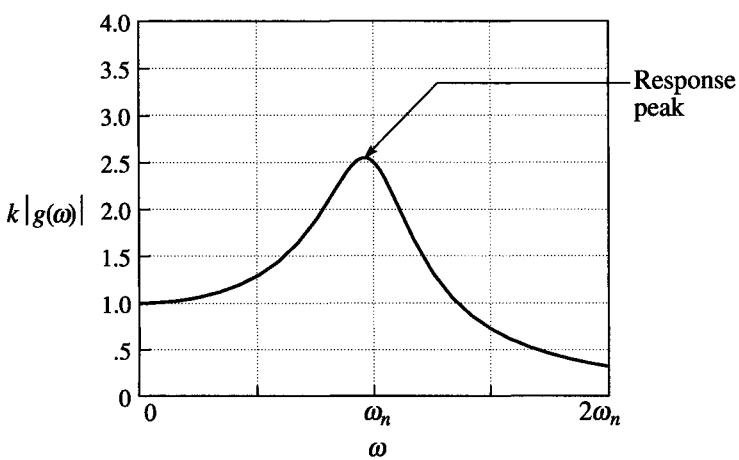


Figure 2.29 Typical underdamped response showing a peaked response

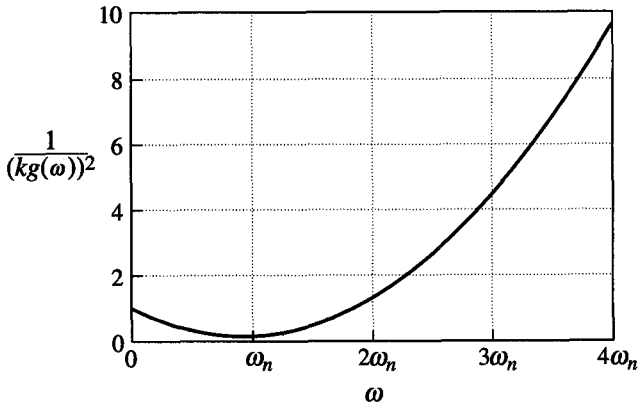


Figure 2.30 Graph with minimum corresponding to the maximum amplitude response

All these observations are reflected in (2.10.12). Clearly, as ζ is increased, ω_p drops. Furthermore, when $\zeta = \frac{1}{\sqrt{2}}$, $\omega_p = 0$. Thus the response peak occurs at $\omega = 0$. For $\zeta > \frac{1}{\sqrt{2}}$, no peak occurs and the response amplitude decays monotonically with increasing frequency.

Next, consider the ratio of the peak response amplitude to that at zero frequency $\frac{|g(\omega_p)|}{|g(0)|}$. Recalling that the response amplitude for a direct force excited system (2.6.28), repeated here, is

$$|x| = \frac{\bar{f}}{m} \frac{1}{\sqrt{(\omega_n^2 - \omega^2)^2 + (2\zeta\omega_n\omega)^2}} \tag{2.10.13}$$

we see that this ratio is equal to

$$\frac{\dfrac{\bar{f}}{m} \dfrac{1}{\sqrt{(\omega_n^2 - \omega_p^2)^2 + (2\zeta\omega_n\omega_p)^2}}}{\dfrac{\bar{f}}{m\omega_n^2}} \tag{2.10.14}$$

We've already found that $\omega_p = \omega_n\sqrt{1 - 2\zeta^2}$, and if we substitute this into (2.10.14) and simplify, we'll obtain

$$\frac{|g(\omega_p)|}{|g(0)|} = \frac{1}{2\zeta\sqrt{1 - \zeta^2}} \tag{2.10.15}$$

Since $\zeta^2 \ll 1$ for systems in which the response is appreciably peaked at ω_p, we can neglect the ζ^2 term and finally obtain

$$\frac{|g(\omega_p)|}{|g(0)|} = \frac{1}{2\zeta} \tag{2.10.16}$$

Now this is a nifty result. It tells us that if we've got the transfer function for our system, we can determine the damping factor just by finding the peak response and dividing by the zero frequency (or static) response.

The final way of finding the damping factor is to look at the *half-power points* of the response curve. By determining the frequencies at which the amplitude of the response drops by a factor of $\sqrt{2}$ from its peak value, we can determine the damping factor. It certainly isn't obvious why this should be so, but it is, as we'll now demonstrate.

We've already determined that the peak response of a mass forced system occurs at $\omega_p = \omega_n\sqrt{1-2\zeta^2}$. What we now want is to determine the frequencies, ω_1 and ω_2, at which the response amplitude is equal to $\frac{1}{\sqrt{2}}\left|g(\omega_p)\right|$. We'll call our unknown half-power frequencies ω_h. Using our previous results, we see that we'll need to solve

$$\frac{\bar{f}}{m}\frac{1}{\sqrt{(\omega_n^2-\omega_h^2)^2+(2\zeta\omega_n\omega_h)^2}} = \frac{\bar{f}}{m}\frac{1}{2\omega_n^2\zeta\sqrt{(2-2\zeta^2)}} \tag{2.10.17}$$

The term on the left is the general expression for the amplitude at a given frequency (2.6.28) while the term on the right is equal to $\frac{1}{\sqrt{2}}$ times the peak response amplitude (from 2.10.15).

To solve this equation we can square both sides, cross-multiply (to get everything in the numerator), and end up with the equation

$$\omega_h^4 + \omega_h^2(4\zeta^2\omega_n^2 - 2\omega_n^2) + \omega_n^4 - 8\omega_n^4\zeta^2(1-\zeta^2) = 0 \tag{2.10.18}$$

If we solve this quadratic expression, we'll get two values for ω_h^2

$$\omega_{h_1}^2 = \omega_n^2(1-2\zeta^2) - 2\omega_n^2\zeta\sqrt{1-\zeta^2} \tag{2.10.19}$$

and

$$\omega_{h_2}^2 = \omega_n^2(1-2\zeta^2) + 2\omega_n^2\zeta\sqrt{1-\zeta^2} \tag{2.10.20}$$

Once we take their square roots, we can subtract ω_{h_1} from ω_{h_2}. If we also neglect terms that involve ζ^2 or higher powers (since they're negligible compared to ζ terms), we finally obtain

$$\omega_{h_2} - \omega_{h_1} = 2\omega_n\zeta \tag{2.10.21}$$

Defining δ_h as

$$\delta_h \equiv \omega_{h_2} - \omega_{h_1} \tag{2.10.22}$$

we can write

$$\zeta = \frac{\delta_h}{2\omega_n} \tag{2.10.23}$$

Here is yet another way to determine ζ. For this identification, you need to find the frequency of the peak response, find the frequencies at which the response has dropped by a factor of $\sqrt{2}$, take their difference, and divide by $2\omega_n$ to obtain the damping factor. All the relevant quantities

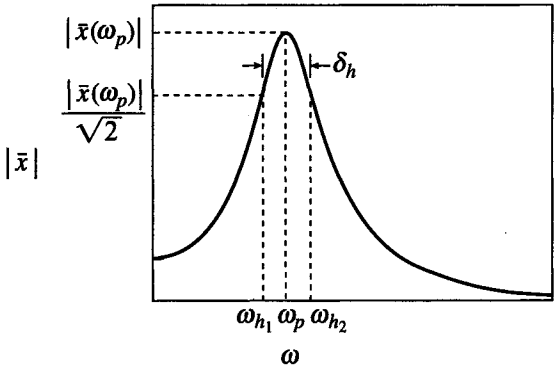


Figure 2.31 Half power points for a sinusoidally forced system

are shown in Figure 2.31. Note that this method is tougher to accomplish in actual practice than the peak response method because the slope of the response curve is quite steep at the half-power points when the damping is low. Thus it's going to be hard to find the half-power frequencies very accurately.

Example 2.15

Problem In this example we'll try to determine the damping factor of a spring-mass-damper system under direct force excitation through use of the half-power points. Figure 2.32 shows the absolute value of the transfer function as a function of frequency. Calculate the desired parameter values.

Solution We know that the magnitude of the transfer function for a direct force excited system is given by

$$|g(\omega)| = \frac{\bar{f}}{m} \frac{1}{\sqrt{(\omega_n^2 - \omega^2)^2 + (2\zeta\omega\omega_n)^2}}$$

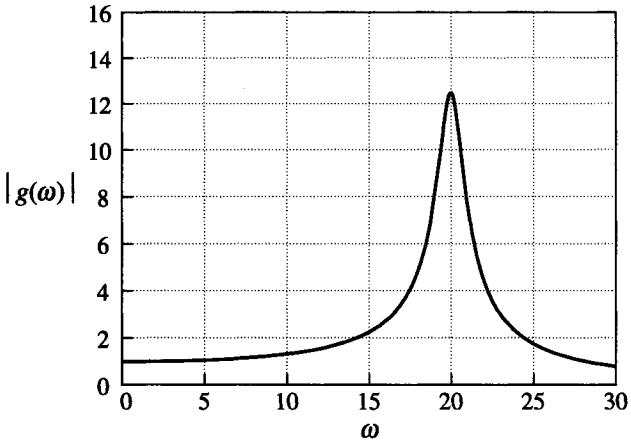


Figure 2.32 Absolute value of transfer function for underdamped spring-mass-damper system

We also know that the damping factor is approximated by $\frac{\delta_h}{2\omega_n}$ where δ_h is the difference in frequency of the two half-power points (review Figure 2.31).

From the graph we can estimate that $|g|_{max} = 12.5$ and occurs at 20 rad/s. The half-power points correspond to an amplitude of $\frac{12.5}{\sqrt{2}} = 8.84$. The frequencies corresponding to this are around 20.8 and 19.1. Thus we have $\delta_h = 20.8 - 19.1 = 1.7$ rad/s. We can then find ζ from the half-power formula:

$$\zeta = \frac{1.7}{(2)(20)} = .0425$$

We can validate this result by using the ratio of $|g_{max}|$ to $|g_0|$

$$\frac{1}{2\zeta} = \frac{|g_{max}|}{|g_0|} = \frac{12.5}{1} = 12.5$$

Solving for ζ gives us $\zeta = .04$, which is in close agreement with the damping factor value we found from the half-power analysis.

Just for interest, the values used to generate Figure 2.32 were $\bar{f} = 400$ N, $m = 1$ kg, $\omega_n = 20$ rad/s, and $\zeta = .04$.

2.11 OTHER TYPES OF DAMPING

It's already been mentioned that viscous damping, although convenient from a mathematical point of view, doesn't really model the damping found in actual structures very well. In this section we'll look other damping models and see how their inclusion affects the response of our systems.

The first type of damping we'll look at is dry friction (also called Coulomb damping). You've undoubtedly already seen this type of damping in your physics and dynamics classes. The simplest dry friction model assumes that two contacting bodies that are moving relative to each other will experience a constant friction force that always acts to oppose the motion. Figure 2.33 illustrates this behavior. The magnitude of the force is equal to μW, where W is the weight of the object that is sliding and μ is the coefficient of friction between the object and the surface it's sliding on. This force can be represented mathematically as

$$F_d = \mu W \text{sign}(\dot{x}) \tag{2.11.1}$$

where $\dot{x}$ is the relative velocity and sign( ) indicates the sign function (equal to 1.0 when the argument is greater than zero, zero at zero, and -1.0 when the argument is less than zero).

If we use this sort of damping with our spring-mass system, our equation of motion will be

$$m\ddot{x} + d\,\text{sign}(\dot{x}) + kx = 0 \tag{2.11.2}$$

where for simplicity, we've used d to stand for μW.

The free vibration behavior of this system is rather interesting. Since the damping is constant (as long as the velocity's sign doesn't change), we can rewrite (2.11.2) as follows:

$$\begin{aligned} m\ddot{x} + kx &= -d, \qquad \dot{x} > 0 \\ m\ddot{x} + kx &= d, \qquad \dot{x} \le 0 \end{aligned} \tag{2.11.3}$$

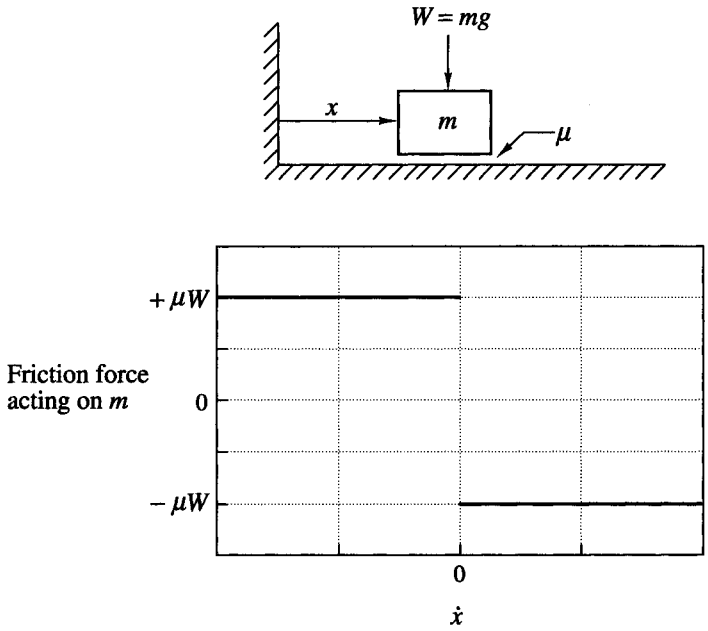


Figure 2.33 Force versus velocity relationship for dry friction

What this tells us is that we can consider the system to be a piecewise linear, forced system; i.e., the system looks like a linear spring-mass system being driven by a constant force. The only difference between this system and a completely linear one is that the forcing changes sign when the velocity changes sign. If we give this system an initial displacement and then release it, the envelope of the resulting oscillations is itself linear. This linear decay of amplitude is quite different from what we saw with linear damping, which produced an exponentially decaying envelope for the oscillations. A typical plot of a dry friction damped response is shown in Figure 2.34. Note that the oscillations stop in a finite time, whereas in the case of linear damping the oscillations continue forever (although they get very small). Also note that the response stops at a nonzero value. This is because the oscillations eventually become so small that the spring force is unable to overcome the dry friction force and the mass gets "stuck" at some deflected position. The precise position will depend upon the initial conditions.

A dry friction model is often used to approximate the damping that occurs when parts rub against each other. Thus if you were trying to estimate the damping that might exist in a large space structure, something made up of many truss elements, you might try to estimate the dry frictional losses due to relative motion between the truss elements and the joint assemblies that hold them together.

Example 2.16

Problem From the response of a spring-mass system with a dry friction interface in Figure 2.34, determine the linear damping estimates that come from examining the change in the peak response between peaks 1 and 2, peaks 2 and 3, and peaks 1 and 3. Is the linear estimate found from two

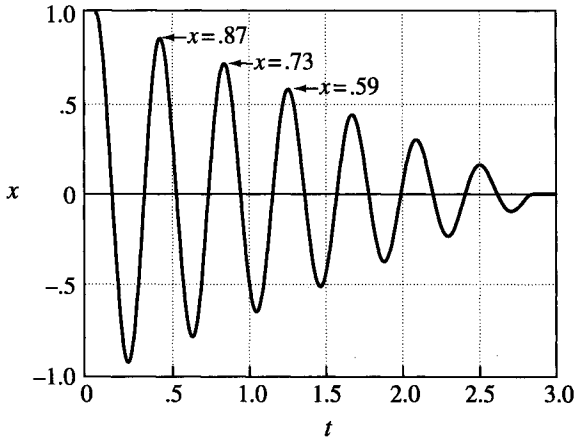


Figure 2.34 Time response of a dry friction damped system

oscillation cycles (peak 1 to peak 3) equal to the average of that found for the peaks 1-2 and peaks 2-3 estimates?

Solution We saw in (2.10.5) that

$$\zeta = \frac{\sigma}{\sqrt{4\pi^2 + \sigma^2}}$$

and in (2.10.10) that

$$\sigma = \frac{1}{n}\ln\left(\frac{x(0)}{x(nt_p)}\right)$$

Using the data from Figure 2.34 we see that

$$\sigma_{1-2} = \ln\left(\frac{.87}{.73}\right) = .175$$

and therefore

$$\zeta_{1-2} = .028$$

Looking between peak 2 and peak 3 gives us

$$\sigma_{2-3} = \ln\left(\frac{.73}{.59}\right) = .213$$

Thus we have

$$\zeta_{2-3} = .034$$

Last, the log decrement between the first and third peaks is

$$\sigma_{1-3} = \ln\left(\frac{.87}{.59}\right) = .388$$

and the associated linear damping estimate is

$$\zeta_{1-3} = .062$$

We can immediately see that the damping couldn't have been linear in the original system. If the damping had been linear, all the linear damping estimates would have been identical (assuming no experimental error), as would all the log decrements. What these results show is that the log decrement over two cycles is equal to the sum of the log decrements from peak 1 to 2 and from peak 2 to 3. Furthermore, the linear damping estimate from peaks 1 to 3 is equal to the sum of those from peak 1 to 2 and from peak 2 to 3. Thus we now see what kind of trend should tip us off to the presence of dry friction. If the damping estimates increase as time increases (as ours did, going from .028 to .034), then we may have dry friction. If the overall damping estimate between three peaks is equal to the sum of those found from the two adjacent peaks, then we have even more reason to suspect dry friction. Of course, we might have a combination of linear and dry friction, and so we'd have to try to identify the amount of each particular component.

The other kind of damping we'll consider is called *structural* or *hysteretic* damping. This kind of damping does a reasonable job of approximating the kind of energy loss experienced by materials that are repeatedly loaded and unloaded. In testing for this type of damping, a specimen is cyclically loaded and the testing machine records the stress developed during the loading. If the system has internal damping, the response of such a test will look like curve shown in Figure 2.35. When the damping test is run, the curve is traversed in a clockwise manner. After one full cycle of testing, the material's state is back where it started. However the fact that there is a finite area enclosed by the curve implies that energy was lost during the test. If no damping had been present, the system would have unloaded along the same path that it took when loading was applied. When materials unload along a different path (as illustrated), the effect is called hysteresis, which tells us why this damping is also called hysteretic damping.

You can see that the area within the stress-strain curve is proportional to energy loss when you realize that stress is a force quantity and strain is a displacement quantity. Thus the integral represents a force times displacement quantity, i.e., a work term. The area within the curve is proportional to the work done on the specimen and indicates how much energy was dissipated by the specimen during the test.

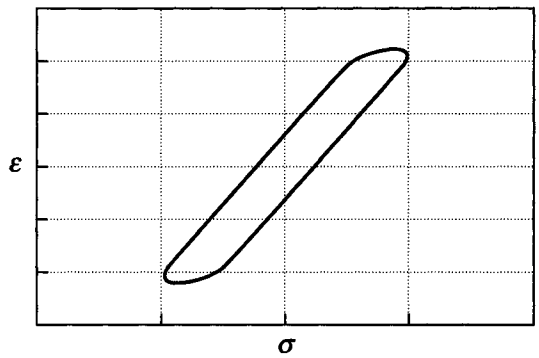


Figure 2.35 Stress-strain curve for hysteretically damped material

It turns out that the stress-strain curve doesn't depend upon frequency (at least within a reasonable range of frequencies). You can also see that doubling the amplitude of the strain will double the entire curve (both horizontally and vertically). The result of this is that the energy dissipated over one cycle will be proportional to the square of the amplitude of the motion. (The energy is proportional to the area enclosed in the curve and the area is proportional to the square of the amplitude.)

Next we need to find out how we can deal with this kind of damping. We'd like to come up with the equivalent viscous damping that will match the hysteretic damping in some sense. The most convenient approach to this question is to try to get the viscous damping to dissipate the same amount of energy over one cycle as the hysteretically damped material would. Therefore, our immediate task is to find out how much energy is dissipated over a cycle by a viscous damper.

As usual, we'll use complex notation to obtain our solution. We can assume that the forcing is actually equal to $\bar{f}\cos(\omega t)$ (with $\bar{f}$ real), and thus our solution is given by the real part of x that results from solving

$$m\ddot{x} + c\dot{x} + kx = \bar{f}e^{i\omega t} \tag{2.11.4}$$

We've solved this problem earlier in the chapter, and the complete solution is given by

$$x(t) = \bar{f}|g(\omega)|e^{i(\omega t-\phi)} \tag{2.11.5}$$

where $|g|$ and ϕ are defined in (2.7.8) and (2.7.9). The velocity is found by differentiating the displacement response, yielding

$$\dot{x}(t) = i\omega\bar{f}|g(\omega)|e^{i(\omega t-\phi)} \tag{2.11.6}$$

Now that we have expressions for the applied force and the velocity, we can determine the energy dissipated over one cycle. The work done over one cycle is given by

$$\Delta w = \int f dx \tag{2.11.7}$$

where the integral is taken over the entire displacement history. We can reexpress this in terms of time by realizing that $dx = \dot{x}\,dt$. Thus our work equation becomes

$$\Delta w = \int_0^{\frac{2\pi}{\omega}} f\dot{x}\,dt \tag{2.11.8}$$

Combining (2.11.6) and (2.11.8) gives us

$$\Delta w_v = \int_0^{2\pi/\omega} \mathrm{Re}(\bar{f}e^{i\omega t}) \cdot \mathrm{Re}(i\omega\bar{f}|g|e^{i(\omega t-\phi)})dt \tag{2.11.9}$$

$$= -\bar{f}^2|g|\omega\int_0^{2\pi/\omega} \cos(\omega t)\sin(\omega t-\phi)dt \tag{2.11.10}$$

$$= \bar{f}^2|g|\pi\sin(\phi) \tag{2.11.11}$$

where Re( ) indicates that the real part of the argument is being considered and w_v indicates that the energy dissipated is due to viscous damping. Recalling (2.7.9), we can find $\sin(\phi)$ to be

$$\sin(\phi) = \frac{2\zeta\omega\omega_n}{\sqrt{(\omega_n^2 - \omega^2)^2 + (2\zeta\omega\omega_n)^2}} \tag{2.11.12}$$

Comparing this with (2.7.8) and recalling that $2\zeta\omega_n = \frac{c}{m}$ gives us

$$\sin(\phi) = 2\zeta\omega\omega_n m|g| \tag{2.11.13}$$

Thus (2.11.11) becomes

$$\Delta w_v = (\bar{f}|g|)^2 \pi c\omega \tag{2.11.14}$$

Finally, $\bar{f}|g|$ is simply the magnitude of the resultant oscillations $|\bar{x}|$. Thus we have

$$\Delta w_v = |\bar{x}|^2 \pi c\omega \tag{2.11.15}$$

This result tells us that the energy absorbed because of viscous damping over one cycle of loading is proportional to the square of the amplitude of the response ($|\bar{x}|^2$), and it varies linearly with the damping coefficient c and with the frequency ω of the test. Thus, if we double the frequency, we'll double the energy absorbed. This makes sense because the force of a viscous damper is proportional to velocity, and doubling ω will double the velocity. It's also logical that the amount of damping will vary with the damping coefficient. You'd expect that doubling the damping coefficient should double the energy absorbed.

Now that we've seen how energy is consumed by a viscous damper, we can go ahead and determine what an equivalent viscous damper might be for hysteretic damping. We've already mentioned that the energy absorbed by a hysteretic damper is proportional to the square of the oscillation amplitude and isn't proportional to the forcing frequency, while we've just seen that the energy absorbed by a viscous damper has a frequency dependence. If the damping force generated by the viscous damper had been proportional to displacement instead of velocity, then we wouldn't have had a frequency dependence. However, a force that's simply proportional to displacement is a spring force and doesn't dissipate energy. It's necessary for the force to be out of phase with respect to the displacement for energy dissipation to occur. Realizing this, let's assume that hysteretic damping is proportional to displacement but is out of phase with it. We could do this by using complex exponentials as our solutions and assuming that the force developed is proportional to ix (with the spring force proportional to x and a viscous damper's force proportional to $i\omega x$). Of course, this would be tough to justify on physical grounds. However, let's run with this idea anyway and see where it leads.

If we take the damping force to be $ik\gamma x$, our equation of motion becomes

$$m\ddot{x} + kx + ik\gamma x = f \tag{2.11.16}$$

Since both spring and damper are proportional to displacement, we can combine them thus:

$$m\ddot{x} + (1 + i\gamma)kx = f \tag{2.11.17}$$

$k(1 + i\gamma)$ is called the *complex stiffness* and γ is the *structural damping factor.*

We can quickly go through the same sort of analysis we've just finished for the viscous case to find how much energy is dissipated by hysteretic damping. If we let $f = \bar{f}e^{i\omega t}$ (with $\bar{f}$ real) and $x = \bar{x}e^{i\omega t}$, we can solve (2.11.17) to find

$$\bar{x} = \frac{\bar{f}}{k - \omega^2 m + i\gamma k} \tag{2.11.18}$$

Multiplying the real part of the force and the real part of the velocity, we can form the dissipated energy

$$\Delta w_s = \int_0^{2\pi/\omega} \bar{f}\cos(\omega t)\left(-\omega\bar{f}\right)\frac{(k - \omega^2 m)\sin(\omega t) - \gamma k\cos(\omega t)}{(k - \omega^2 m)^2 + (\gamma k)^2}\,dt \tag{2.11.19}$$

Evaluating this gives us

$$\Delta w_s = \frac{\pi\gamma k\bar{f}^2}{(k - \omega^2 m)^2 + (\gamma k)^2} \tag{2.11.20}$$

Finally, recognizing from (2.11.18) that $|\bar{x}|^2$ shows up in our work expression, we can form the final result

$$\Delta w_s = \pi\gamma k|\bar{x}|^2 \tag{2.11.21}$$

This is exactly what we wanted, a dissipation that's proportional to the square of the amplitude and has no frequency dependence. It gives us a conceptual model for structural damping, i.e., a damping force that varies with displacement and is 90 degrees out of phase with it. If we compare the energy dissipated by viscous and structural damping

$$\Delta w_v = c\pi\omega|\bar{x}|^2$$

$$\Delta w_s = \gamma k\pi|\bar{x}|^2$$

we see that we can come up with an equivalent viscous damping coefficient c_{eq} by letting

$$c_{\text{eq}}\pi\omega|\bar{x}|^2 = \gamma k\pi|\bar{x}|^2 \tag{2.11.22}$$

or

$$c_{\text{eq}} = \frac{\gamma k}{\omega} \tag{2.11.23}$$

How would you go about actually determining c_{eq}? First you'd have to run a stress-strain test to determine what the energy absorption is for your material. Since you'll know the amplitude of your displacement during the test, you'll be able to use (2.11.21) to determine γ. Then you'd use (2.11.23) to calculate c_{eq}. You'll note that you also need ω. This implies that the c_{eq} will depend on the frequency at which your system will be oscillating. This is generally okay, since there's usually a primary vibration going on. It's important, however, to realize that your equivalent viscous coefficient is good only at this particular frequency. If the frequency changes, you have to recalculate c_{eq}.

A related observation is that these damping analyses are applicable only to a system that's vibrating in a sinusoidal way. It would be nice if we could also determine the response of the system to a nonsinusoidal input (a step or impulse), as we can with viscous damping. Unfortunately, that's not possible without changing our damping model drastically and increasing the complexity of the analysis by a very significant amount.

Example 2.17

Problem A sinusoidal stress-strain test has been run on a system and you're to determine the structural damping factor. The linear stiffness k was 100,000 N/m, the driving force had an amplitude of 350 N, and the amplitude of the response was .01 m. Furthermore, the output differed from the input by 90 degrees.

Solution An examination of (2.11.18) shows that the output and input differ by 90 degrees when $\omega = \sqrt{\frac{k}{m}}$. For this situation we have

$$|\bar{x}| = \frac{\bar{f}}{|\gamma k|}$$

Using the given data, this means that

$$.01 = \frac{350}{100,000\gamma}$$

which gives us $\gamma = .35$.

Example 2.18

Problem For the system discussed in Example 2.17, determine the energy dissipated during one cycle of the stress-strain test.

Solution The energy dissipated is given by

$$\Delta w_s = \pi \gamma k |\bar{x}|^2$$

Since we've already determined the value for γ and are given k and $|\bar{x}|$, we can immediately solve for the energy dissipated:

$$\Delta w_s = \pi(.35)(100,000)(.01)^2 = 11 \text{ N·m}$$

2.12 ACCELEROMETERS AND SEISMOMETERS

Transducer is the word given to describe the devices that allow us to measure the relevant variables in a vibrations experiment. One of the early decisions that must be made in the course of a vibrations test is the determination of which transducer is appropriate for the particular problem at hand. In this section, we'll discuss two kinds of transducer and give some insight into their uses and limitations.

Perhaps the most widely used vibrational transducer is the *accelerometer*. Although there's more than one way to build an accelerometer, the most common is the piezoelectric type, shown for

two configurations in Figure 2.36. The basic mechanism at work for both is a mass that acts on a piezoelectric crystal. Because piezomaterials generate an electric charge that's proportional to their deformation, we can use them to determine the acceleration of the object on which the accelerometer is attached. In Figure 2.31*a* the piezoelectric material is sheared as the seismic mass moves relative to the surrounding casing, thus generating a charge. The accelerometer shown in Figure 2.31*b* induces a charge in the piezoelectric material by compressing it.

Both the configurations of Figure 2.36 can be viewed as simple SDOF systems (a mass attached to the accelerometer casing through a spring). Referring to Figure 2.37, we see that the internal movable mass within the accelerometer body is denoted by m, the spring that connects this mass to the accelerometer is denoted by k, the actual motion of the accelerometer body is given by y, and the relative motion of the mass m with respect to the accelerometer body is given by x. We've also included a damper c to account for frictional dissipation. It is very important to understand that x is the *relative* motion of the internal mass, not its actual motion with respect to an outside observer.

Remembering that the absolute motion of the seismic mass is given by $x + y$, we can perform a force balance to find

$$m(\ddot{x} + \ddot{y}) = -kx - c\dot{x} \tag{2.12.1}$$

Rearranging this yields

$$m\ddot{x} + c\dot{x} + kx = -m\ddot{y} \tag{2.12.2}$$

and dividing by m brings us to

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2 x = -\ddot{y} \tag{2.12.3}$$

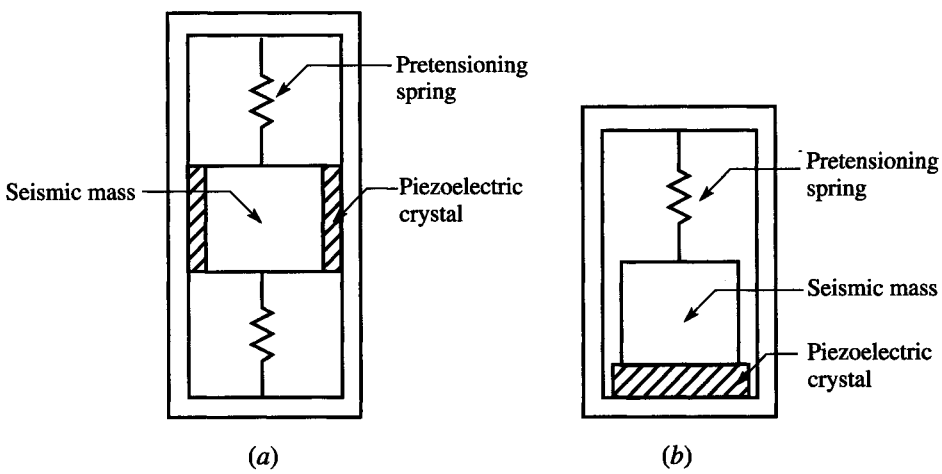


Figure 2.36 Accelerometer configurations

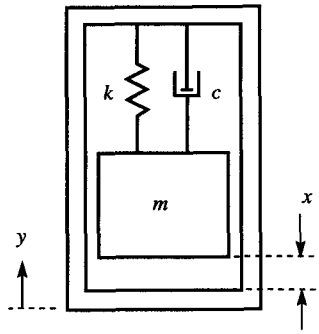


Figure 2.37 Simple accelerometer model

If we assume that the accelerometer body is oscillating sinusoidally, then using $y(t) = \bar{y}e^{i\omega t}$ and $x(t) = \bar{x}e^{i\omega t}$ will lead to

$$\left(\omega_n^2 - \omega^2 + 2i\zeta\omega_n\omega\right)\bar{x} = \omega^2\bar{y} \tag{2.12.4}$$

Therefore we have

$$x(t) = \frac{\omega^2\bar{y}e^{i\omega t}}{\omega_n^2 - \omega^2 + 2i\zeta\omega_n\omega} \tag{2.12.5}$$

This is the govening equation for our accelerometer. Although this doesn't look particularly simple, this situation changes when we look at a limiting case for which the excitation frequency is much smaller than the natural frequency of the accelerometer, i.e., $\omega \ll \omega_n$. In this case the velocity and acceleration terms become negligible in comparison to displacement ($|\bar{x}| \gg |i\omega\bar{x}| \gg |-\omega^2\bar{x}|$). Therefore (2.12.3) can be simplified to

$$x(t) = -\frac{1}{\omega_n^2}\ddot{y}(t) \tag{2.12.6}$$

Thus we have a direct measure of the object's acceleration: we need simply multiply $x(t)$ by $-\omega_n^2$ to obtain it.

This simple formula holds only as long as $\omega \ll \omega_n$, thus telling us the limitation on the usable frequency range.

In addition to being used as an accelerometer, the setup can be used to obtain a different kind of measurement, namely, the displacement of the mounting surface. Consider the case in which the natural frequency of our device is now much lower than the exciting frequency. This is the situation that corresponds to point 5 on our response graph of a seismically excited mass (as shown previously in Figure 2.7). You'll recall that in this case your hand moved up and down while the mass stayed fixed. Thus the relative motion between the mass and your hand was exactly the same as the actual motion of your hand, just 180 degrees out of phase. This is the result we should expect the math in this case to give us.

If $\omega \gg \omega_n$ in (2.12.5) we obtain

$$x(t) = -\bar{y}e^{i\omega t} \tag{2.12.7}$$

Thus, as expected, our transducer gives us a reading that's directly proportional to the mounting surface's motion. When used in this manner, the transducer is called a *seismometer*.

Physically, the constraint $\omega \gg \omega_n$ means that the natural frequency of the seismometer must be small, thus implying either a large mass or a weak spring. Consequently, seismometers are usually substantially larger than accelerometers.

Example 2.19

Problem Assume a given accelerometer has a moving mass of 10 g and a spring constant of 3.95×10^7 N/m. We'll assume that there is no damping present. What is the upper usable frequency range for this accelerometer if the maximal error that we'll allow is 5%?

Solution The general solution for an accelerometer was given by (2.12.5):

$$x(t) = \frac{\omega^2 \bar{y}e^{i\omega t}}{\omega_n^2 - \omega^2 + 2i\zeta\omega_n\omega}$$

We can recognize the numerator as simply $-1\times$ the acceleration of the base. Thus the $-x$ response to acceleration ratio is given by

$$-\frac{x(t)}{\ddot{y}} = \frac{1}{\omega_n^2(1 - \frac{\omega^2}{\omega_n^2})}$$

As the frequency of excitation increases, the ratio of x to acceleration grows. Therefore a 5% error means we'll have

$$1.05 = \frac{1}{1 - \frac{\omega^2}{\omega_n^2}}$$

or

$$\frac{\omega}{\omega_n} = .2182$$

Since $\omega_n = 62,850$ rad/s (using the given values for k and m) this means $\omega = 13,700$ rad/s or 2180 Hz. Thus the maximum usable frequency range is about 20% of the system's natural frequency.

2.13 HOMEWORK PROBLEMS

Sections 2.2 and 2.3

2.1. An existing damped, SDOF system like the one shown in Figure P2.1 is too lightly damped. In the original system $m = 1$ kg, $k = 10,000$ N/m, and $c = 15.0$ N·s/m. To reduce the setting time by 70%, what must c be increased to?

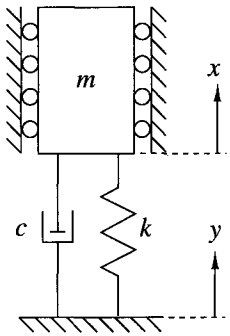


Figure P2.1

2.2. What will doubling the mass in the lightly damped SDOF system (of Figure P2.1) do to the settling time?

2.3. If k is doubled in the lightly damped SDOF system of Figure P2.1, what happens to the settling time?

2.4. We're going to analyze the motions of a unicyclist. The actual system is shown in Figure P2.4, as is our approximate model. If the unicyclist is traveling at 3 m/s and hits the illustrated bump, what will the cyclist's maximum deflection from x_{eq} be?

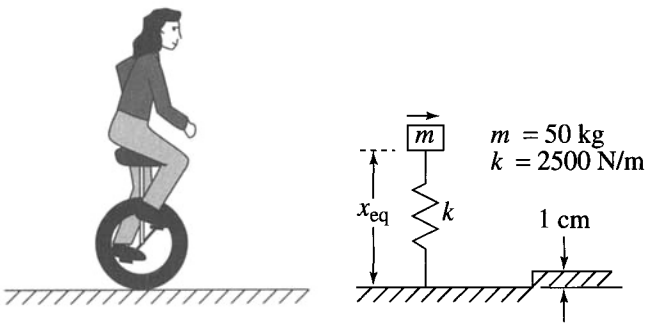


Figure P2.4

2.5. Consider the unicyclist of Problem 2.4. The road she's traveling on has a sinusoidal profile, as illustrated in Figure P2.5. What is the worst speed for her to travel at with regard to her vibrational response?

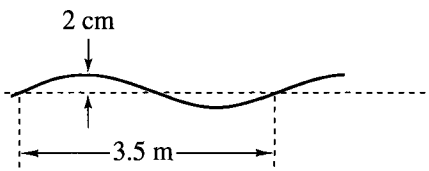


Figure P2.5

2.6. Dopey Dog, an animated film star, has just been handed a bone with an attached, stretched, elastic rubber band. After a momentary pause (inserted in violation of realism so the audience can see the amusing moment of shocked realization on Dopey's face) he begins to rocket to the right. If $m_{\text{Dopey}} = 45$ kg, $k_{\text{elastic}} = 300$ N/m, and the elastic is stretched 6 m when handed to Dopey, calculate Dopey's acceleration and velocity at $t = .3$ second (ignore the unstretched length of the rubber band, i.e., set it to zero).

2.7. A machine on its suspension deflects .6 cm under static conditions. What is the ratio of the mass's oscillation amplitude to the base oscillation amplitude if the base is oscillating at 40 Hz? Assume zero damping.

2.8. The spring-mass system shown in Figure P2.8 has a spring constant of 1600 N/m and a mass of 2 kg. What is the response amplitude to a ground input of

$$y(t) = .06\sin(10\,t)\ \text{m}$$

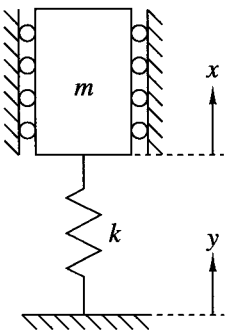


Figure P2.8

2.9. The spring-mass system shown in Figure P2.8 experiences a 2 mm in-phase oscillation when the base is oscillating at 300 rad/s with an amplitude of 1.1 mm. What is the static deflection of the mass?

2.10. The spring-mass system shown in Figure P2.8 experiences a 4 mm out-of- phase oscillation when the base is oscillated at 200 rad/s with an amplitude of 1.3 mm. What is the static deflection of the 5 kg mass?

2.11. Consider again the system shown in Figure P2.8. When excited by a ground motion of $.001\cos(20\,t)$ m, the response has an magnitude of 1.25 mm and is in phase with the excitation. Find the natural frequency of the system.

2.12. Consider once more the system shown in Figure P2.8. This system is excited by a ground motion of $.001\sin(50\,t)$ m. The response is out of phase and has a magnitude of 4 mm. What is the system's natural frequency?

2.13. An 18 kg mass is protected from vibration by being placed in a container and isolated from the walls by two springs. The force characteristic of the first spring is

$$f_1 = 5x(\text{spring } 1)$$

and the other spring can be either of the following:

$$f_2 = x + x^3 \text{ (spring 2)}$$
$$f_3 = 31x \text{(spring 3)}$$

In all three cases the displacement units are assumed to be meters and the forces are newtons. Initially the mass-spring unit is 12 m long. The available space inside the protective box is 8 m. Two possible configurations are being considered, shown in Figure P2.13. Which configuration will be more effective in isolating the mass from external disturbances, assuming that these disturbances occur around $\omega = 3$ rad/s and are of small amplitude ($y(t) = \varepsilon \sin(3.0t)$)? (*Note*: Best isolation here means minimum movement.)

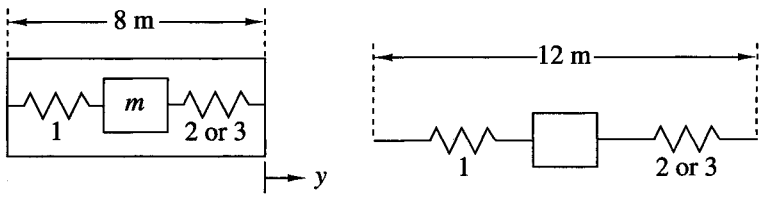


Figure P2.13

2.14. Consider a simple model of a car traveling along a road (the same model as in Problem 2.5). We'll look only at translational motions (vertical oscillations). The road profile is sinusoidal with amplitude .01 m and wavelength 8 m. The speed limit is 35 km/h. Would you experience a smaller oscillation amplitude if you reduced your speed to 25 km/h? $m = 1000$ kg and $k = 70,224$ N/m.

2.15. Consider the system illustrated in Figure P2.15. A rigid ring with mass m is connected via four massless, inextensible wires to a central hub (radius r). Each wire has a tension T. If the motion of the hub is given by $\theta(t) = \theta_0 \cos \omega t$ (θ_0 small), determine the angular response of the ring.

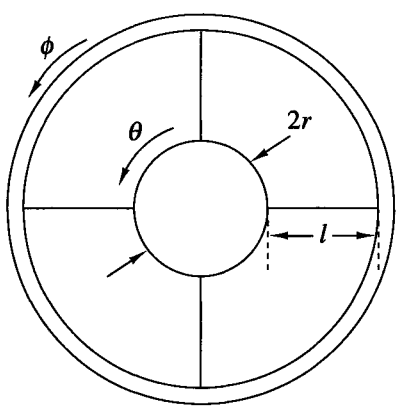


Figure P2.15

2.16. Figure P2.16 shows the steady state time response of an undamped SDOF system under seismic excitation (transients are ignored). Determine ω_n for this system. (*Note*: You *don't* need to try and back out the answer by plugging amplitudes and frequencies into the relevant dynamic equations.)

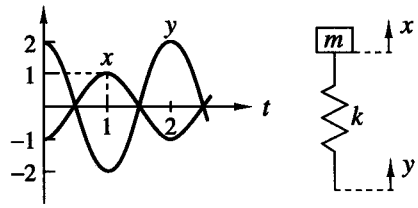


Figure P2.16

2.17. Figure P2.17 shows an enclosure containing a tensioned spring that presses on a mass. The initial compression of the spring is equal to h. The enclosure is subjected to a sinusoidally varying displacement $x(t) = a\cos(\omega t)$. Determine the frequency ω for which the mass will lose contact with the enclosure (consider m, k, h, and a to be fixed, and ignore gravity).

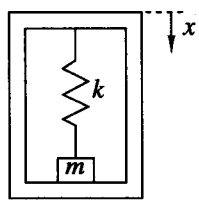


Figure P2.17

2.18. Consider a car that's moving along an undulating road as in Figure P2.18. The road's profile is given by

$$y(x) = .02\sin\left(\frac{2\pi x}{15}\right)$$

where both x and y are in meters. Treat the car as a point mass m. What is the normal force developed between the car and the road as a function of the car's speed $\dot{x}$? ($\dot{x}$ is constant and $m = 1300$ kg)

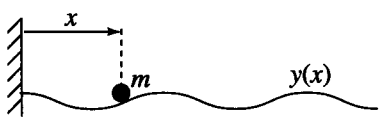


Figure P2.18

2.19. For the system of Problem 2.18, find the maximum and minimum normal force developed between the car and the road if $\dot{x} = 30$ m/s. What is the maximum acceleration that a passenger (m in Figure P2.18) would feel?

2.20. Consider the system of Problem 2.18 but now include the car's suspension, as shown in Figure P2.20. Let $k = 300,000$ N/m, and $m = 1300$ kg. What is the normal force developed between the spring and the road, and what is the maximal acceleration felt by the car? $\dot{x} = 25$ m/s.

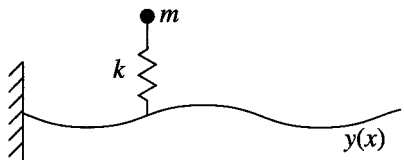


Figure P2.20

2.21. Consider a seismically excited spring-mass system. Find the range of oscillation frequencies ω for which the force transmitted to the mass m is less than or equal to 170 N, with $m = 1$ kg, $k = 25,000$ N/m, and $y(t) = \cos(\omega t)$ (y measured in centimeters).

2.22. Consider an undamped, direct force excited system (Figure P2.22). Let f be zero and let $x = .003\sin(\omega t)$ (measured in meters). What is the force transmitted to the ground? $m = 10$ kg and $k = 5000$ N/m.

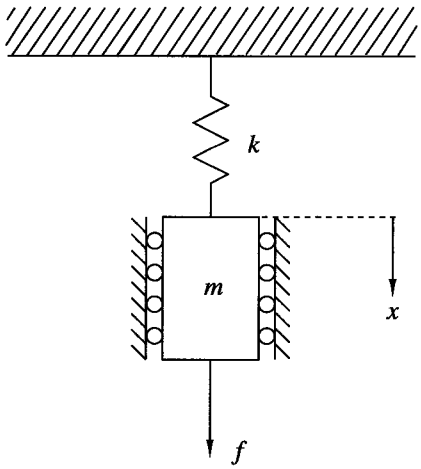


Figure P2.22

2.23. As shown in Figure P2.23, the spring k pushes the mass m against a moving surface. The motion of the surface is given by $x(t) = a(\cos\omega t)$. When $x(t) = 0$, the mass is being pressed into the surface with a force kh (i.e., the spring is precompressed on amount h). Determine the critical speed ω for which the mass will first lose contact with the surface. (*Note*: Identify the critical ω for $a > h$ and $a < h$.)

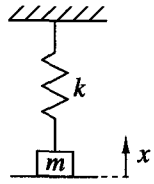


Figure P2.23

2.24. Given the displacement input $y(t) = \bar{y}\cos(\omega t)$, determine the equation of motion for the system illustrated in Figure P2.24 (neglect gravity). What is the natural frequency for the system?

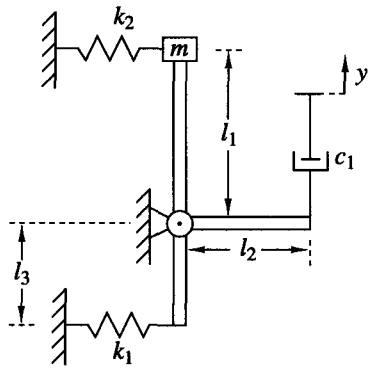


Figure P2.24

Section 2.4

2.25. Using the given data in Figure P2.25, determine the system's natural frequency.

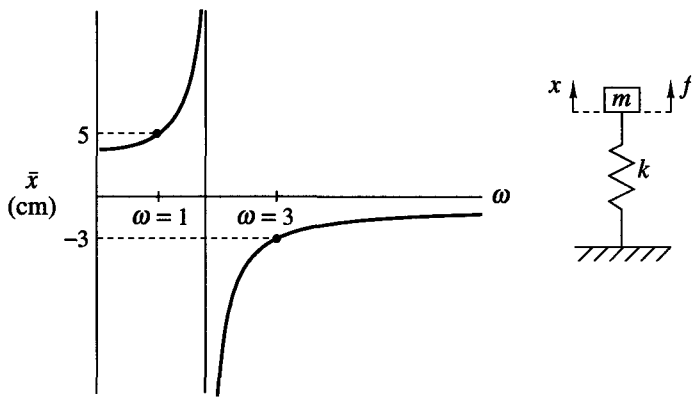


Figure P2.25

2.26. Consider again the undamped, direct force excited system from Figure P2.22. $f(t) = 10\cos(5t)$ N, $m = 10$ kg, $k = 100$ N/m. By how much is the amplitude of the force between the spring and the floor reduced from that applied to the mass?

2.27. Determine the actual response of a spring-mass system when forced at its natural frequency. The relevant equation is

$$\ddot{x} + \omega_n^2 x = \overline{f}\cos(\omega_n t)$$

Assume a solution $x(t) = a_1\cos(\omega_n t) + a_2\sin(\omega_n t) + a_3 t\sin(\omega_n t)$.

2.28. Referring again to the system shown in Figure P2.22, determine the range of frequencies for which m's response amplitude will be below 1 mm if the forcing is equal to $1.1\cos(\omega t)$ N, $m = 10$ kg and $k = 10,000$ N/m.

2.29. Consider once more the system shown in Figure P2.22. Let $m = 5$ kg, $k = 20,000$ N/m, and $f(t) = 20\sin(\omega t)$ N. For what range of ω's will the amplitude of the response be less than 20 mm?

2.30. Consider the two-mass/two-spring system shown in Figure P2.30. m_1 is moved to the right until it contacts m_2. The left end of k_1 is then moved an additional distance $F_0\left(\frac{1}{k_1} + \frac{1}{k_2}\right)$ to the right, causing m_1 and m_2 to press tightly together. Further motions of k_1's left end will be labeled z, motion of m_1 will be labeled x, and motion of m_2 will be labeled y. As long as m_1 and m_2 remain in contact, $x = y$. Derive an expression for the force between the two masses as a function of frequency and show that for an appropriately chosen excitation frequency the masses will separate. Let $z = \frac{F_0}{2k_2}\cos(\omega t)$.

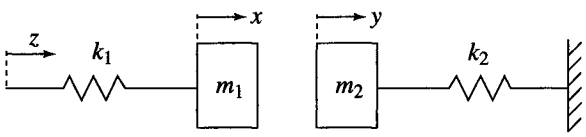


Figure P2.30

2.31. Find the equations of motion and natural frequency for system shown in Figure P2.31 (neglect gravity). What would f need to be for there to be no overall excitation of the system? ($\bar{f}$: newtons, $\bar{y}$: meters). The system is freely hinged at O.

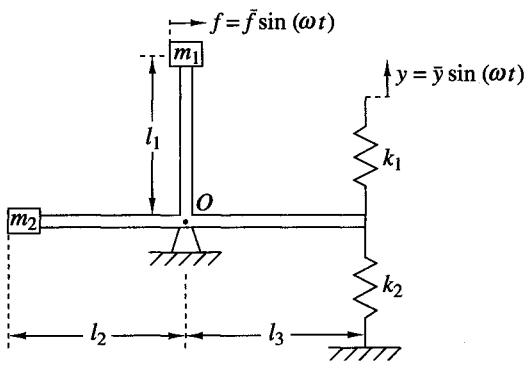


Figure P2.31

2.32. Derive the equations of motion for the system shown in Figure P2.32 for a displacement input given by $y(t) = \bar{y}\cos(\omega t)$ (neglect gravity).

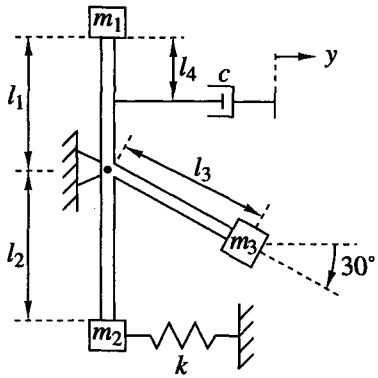
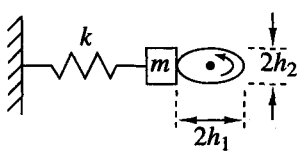


Figure P2.32

2.33. Figure P2.33 is a model of the valves in an engine. m is the mass of a valve, k is the restraining spring, and the rotating body is the cam that drives the valves. Consider the center of this cam to be fixed in space. As the cam rotates, the mass is forced to follow the cam's profile, inducing an x deflection equal to $x_0 + a\cos(\omega t)$. At what rotational velocity do the inertial forces of the piston become so great that the piston loses contact with the cam? When the cam is in the indicated position, the total spring deflection is equal to h_1.



$$x_0 = \frac{h_1 + h_2}{2}$$

$$a = \frac{h_1 - h_2}{2}$$

Figure P2.33

2.34. In Figure P2.34, does the force transmitted to the floor reach a maximum as ω is increased or does it increase indefinitely? At high ω what is more important to the transmitted force, k or c? ($x = x_0 \sin(\omega t)$.)

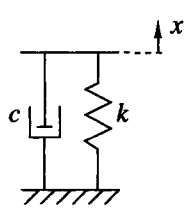


Figure P2.34

2.35. Given the force input $f = \overline{f}\sin(\omega t)$, determine the equation of motion for the system illustrated in Figure P2.35. What is the system's natural frequency? The pivot at O is frictionless. Neglect gravity.

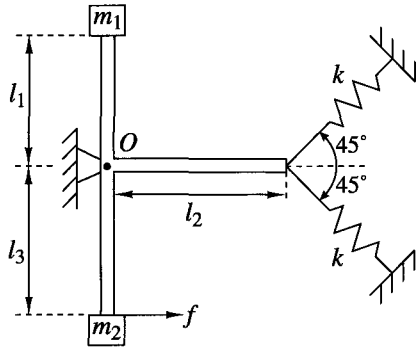


Figure P2.35

Section 2.5

2.36. Determine the transfer function for the system illustrated in Figure P2.36 that relates input force at the mass to output force felt at the floor. ($f = \bar{f}\sin(\omega t)$)

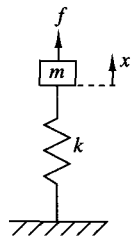


Figure P2.36

2.37. Find the transfer function between m's acceleration and the applied force f for the system shown in Figure P2.36.

2.38. Find the transfer function of acceleration $\ddot{x}$ to position x for the system of Figure P2.36.

2.39. Find the transfer function of support excitation y to response angle θ for the pendular system shown in Figure P2.38. Make sure to linearize your equations. The pendulum is of length l and the freely pivoted upper end of the pendulum is moved horizontally according to

$$y(t) = a\sin(\omega t)$$

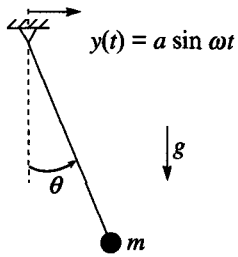


Figure P2.39

2.40. Find the transfer function between the displacement input y and the displacement output x for the system shown in Figure P2.40. $y = \bar{y}\sin(\omega t)$. The rigid bar pivots freely at O.

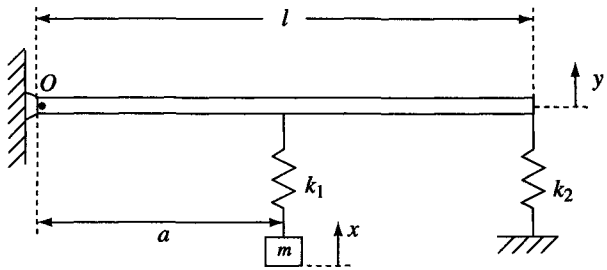


Figure P2.40

2.41. Find the transfer function between the input displacement y and the output force against the wall for the system shown in Figure P2.41. $y = \bar{y}\sin(\omega t)$

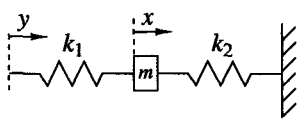


Figure P2.41

2.42. Consider the pendular system illustrated in Figure P2.42 (lumped mass m on the end of a rigid rod of length l). A small motor at O produces a sinusoidally varying torque ($M = \bar{M}\sin(\omega t)$) that acts on the pendulum. Find the transfer function from input torque to response angle $\theta(t)$. Linearize your system equations about $\theta = 0$.

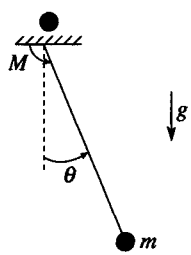


Figure P2.42

Section 2.6

2.43. Consider the spring-mass-damper system illustrated in Figure P2.43. You need to determine the actual values for k and c but have only a limited amount of information. Given the data shown, determine k and c.

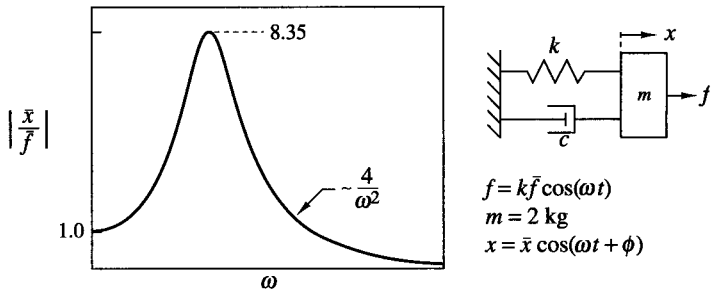


Figure P2.43

2.44. Is the plot illustrated in Figure P2.44 (actual displacement vs. frequency for a direct force excited, damped oscillator) a reasonable one?

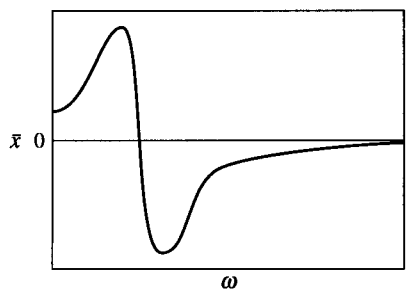


Figure P2.44

2.45. Figure P2.45 shows the steady state response (transients are ignored) of a direct force excited, SDOF system. The mass is equal to 2 kg. What other system parameters can you determine?

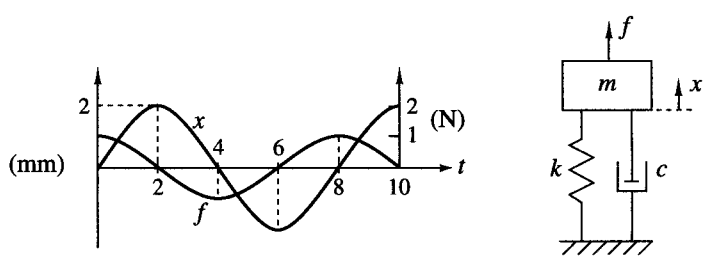


Figure P2.45

2.46. Let $l_1 = 1$ m, $l_2 = .5$ m, $l_3 = 2$ m, $m_1 = 1.1$ kg, $k = 100$ N/m, and $c = 3$ N·s/m. Determine the equation of motion for the system illustrated in Figure P2.46 in terms of the parameters ω_n and ζ. Neglect gravity and assume that the maximal response occurs when the system is forced at $\omega = \omega_n\sqrt{1-2\zeta^2}$.

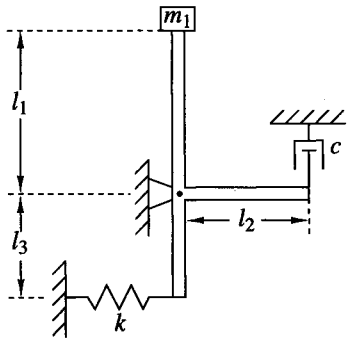


Figure P2.46

Section 2.7

2.47. Put $x(t) = 4\cos(3t) - 2\sin(3t)$ into complex form so that the real part of the complex number gives us $x(t)$.

2.48. Can $x(t) = 3\cos(4t) - 3\sin(2t)$ be put into complex form so that $x(t)$ is expressed as the real part of a single complex term?

2.49. Express $x(t) = \cos(t) - \sin(t)$ in purely complex form, i.e., so that there is no need to take the real or imaginary part of the complex variables.

2.50. Referring again to the system shown in Figure P2.1, find the complex transfer function between the input displacement y and the output force against the ground.

2.51. Find the complex transfer function between the velocity input $\dot{y}$ and the displacement output x for the system illustrated in Figure P2.51.

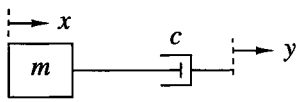


Figure P2.51

2.52. Find the transfer function from input force f to output velocity $\dot{x}$ for the system shown earlier in Figure P2.22. Let $f = \bar{f}e^{i\omega t}$ and $x = \bar{x}e^{i\omega t}$.

2.53. Find the transfer function from the input force f to output acceleration $\ddot{x}$ for the system shown in Figure P2.22. Let $f = \bar{f}e^{i\omega t}$ and $x = \bar{x}e^{i\omega t}$. Express the system equation in terms of ω_n and ζ and use the nondimensionalized frequency Ω.

2.54. What is the real response of $5\ddot{x} + 2\dot{x} + 300x = 4\sin\omega t$?

2.55. Find the real response of $\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2 x = \sin\left(\omega t + \frac{\pi}{4}\right)$

2.56. Solve for the velocity response $\dot{x}$ of the system illustrated in Figure P2.56. $y = .02\sin(150t)$, $k_1 = 10,000$ N/m, $k_2 = 5000$ N/m, $m = .5$ kg, $c = 8.66$ N·s/m.

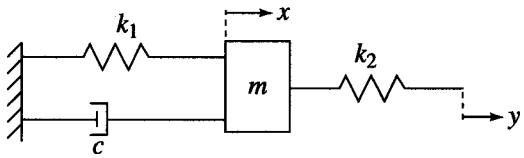


Figure P2.56

2.57. Find the velocity $\dot{x}$ response for the system illustrated in Figure P2.57. $y = .01\sin(100t)$, $k = 8000$ N/m, $c_1 = 4$ N·s/m, $c_2 = 2$ N·s/m, $m = .25$ kg.

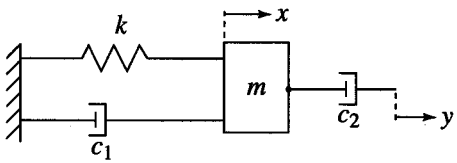


Figure P2.57

Section 2.8

2.58. Add a linear damper in parallel with the spring of the unicycle of Figure P2.5. Let $c = 132$ N·s/m. Under steady state conditions, what will the maximum amplitude response be, and at what velocity will it occur?

2.59. Find the transfer function $\frac{\bar{x}}{\bar{y}}$ for the system illustrated in Figure P2.59. Express the transfer function in both complex and in magnitude/phase form. $m = 10$ kg, $k_1 = 400$ N/m, $c = 14.14$ N·s/m, and $k_2 = 100$ N/m.

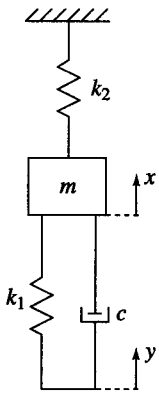


Figure P2.59

2.60. Assume that a seismically excited system is accurately modeled by as shown in Figure P2.1. You need to determine whether the weld connecting the spring to the mass will hold. The maximum stress that the weld can withstand is $1.0 \times 10^6\,\mathrm{N/m^2}$. The mass is 10 kg and the spring constant is 4000 N/m. The damping constant is 40 N·s/m. Is it reasonable to allow the excitation frequency to reach 10 rad/s? The attachment area of the spring is 1 mm × 1 mm and the base amplitude is 1 mm.

2.61. If $y_1 = 5\sin(10t)$, determine what y_2 would have to be for there to be no unbalanced net force acting on m, for the system of Figure P2.61.

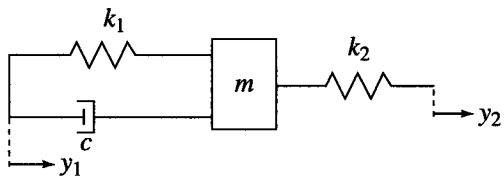


Figure P2.61

2.62. Consider again the seismically excited, damped SDOF system Figure P2.1. What is the magnitude of m's response to a ground excitation of $y(t) = .01\cos(1.25t)$ for $k = 12,000$ N/m, $m = 20$ kg, and $\zeta = .001$? This should give you a general feel for how very lightly damped structures would respond in an earthquake.

2.63. What is the magnitude of the acceleration response for the system shown in Figure P2.1 for $m = 1$ kg, $c = 10$ N·s/m, $k = 100$ N/m, and $y = .005\cos(50t)$?

2.64. Repeat Problem 2.63 and then re-solve for two different cases. In case 1, let $k = 0$ and in case 2, let $c = 0$. Which is dominant, c or k, in determining the actual acceleration response magnitude?

2.65. Repeat Problem 2.62 but use $\zeta = .01$. Does this improve the situation much?

2.66. When the damping is small, increasing the damping by a factor of α will reduce the peak response by the same factor. Does this hold true at large values of damping? Why or why not?

2.67. Show that all the amplitude response plots for varying ζ go through the point $(\frac{\bar{x}}{\bar{y}}) = 1.0$ at $\Omega = \sqrt{2}$ for a seismically excited, damped SDOF system.

Section 2.9

2.68. The plot in Figure P2.68*a* shows amplitude response of the system illustrated in Figure P2.68*b* a centrifuge as a function of frequency *with* a specimen, m_2, in the machine. The mass of the centrifuge is m_1 and it's restrained by a spring of stiffness k. Given that the radius of the spinning chamber is 0.1 m and that the mass m_1 is 100 kg, find the mass of the specimen as well as the spring constant k.

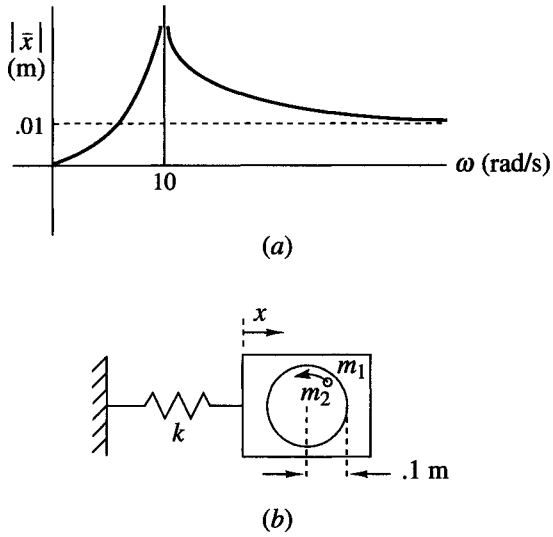


Figure P2.68

2.69. A massless rod (with end masses m_1 and m_2) rotates at an angular speed ω with respect to the massless collar A, which rides along the guide BC, restrained by the two springs, k_1 and k_2. Find the response amplitude of the collar in terms of ω, l_1, l_2, m_1, m_2, k_1 and k_2 in Figure P2.69.

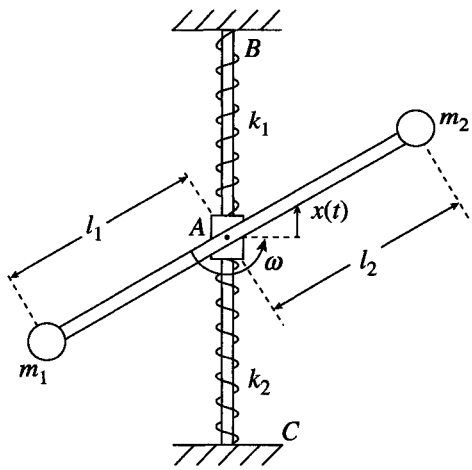


Figure P2.69

2.70. The overhead fan shown in Figure P2.70 isn't dynamically balanced. Because of this, the center of mass of the rotating fan-blade assembly lies off from the center of rotation by a distance l. The rotating mass has a magnitude m_R while the nonrotating motor housing has a mass m_N. Treat the support rod as massless. The radius of gyration of the blade assembly about the point of rotation is equal to k_R. Determine the

fan's response when the blades are rotating at a fixed speed ω. (*Hint*: The spring stiffness in the problem comes from gravity. Look at the fan as a spring-mass system being excited by a rotating mass. Since the angle is small ($\phi \ll 1$), you can look at just the x, y motions.)

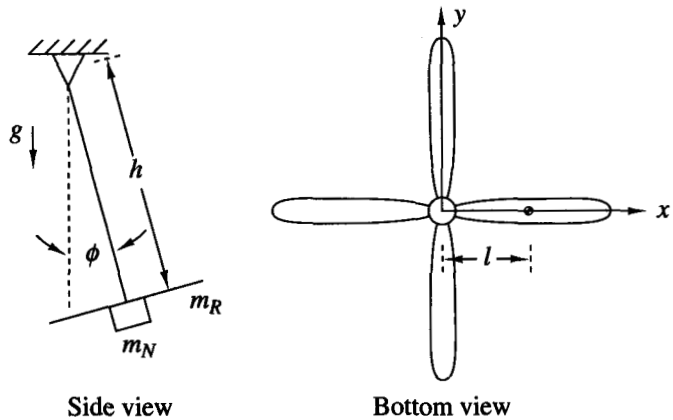


Figure P2.70

2.71. Assume that most of the wet clothes in a dryer (Figure P2.71) are uniformly distributed around the drum (mass = 10 kg). In addition to these clothes, a single .8 kg lump of clothing also lies against the drum's surface. How will these mass terms enter the equation of motion (2.9.5) for the system?

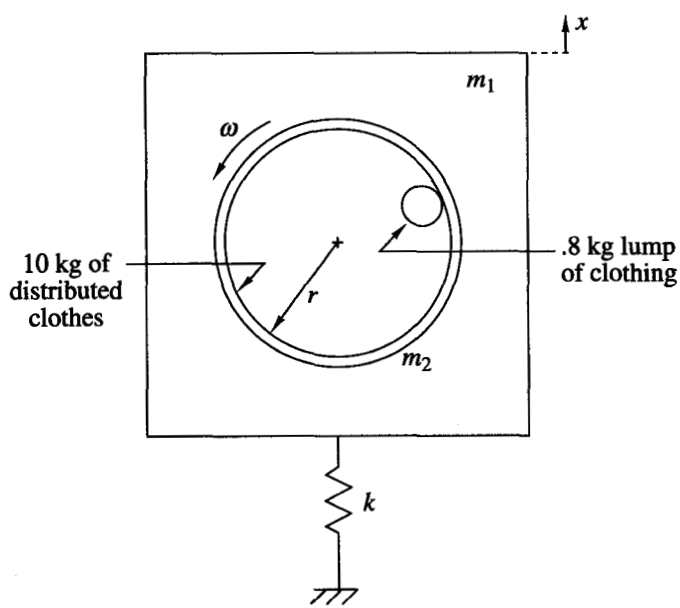


Figure P2.71

2.72. Figure P2.72 illustrates a top view of a top-loading washing machine. The rotational speed of the drum is $\omega = 35$ rad/s and the radius of the drum is $r = .3$ m. Assume that the wet clothes have all gathered into a ball of mass 7 kg (approximated by a point mass). Determine the lateral forces generated by the washing machine. Assume that the mass of the drum is 4 kg.

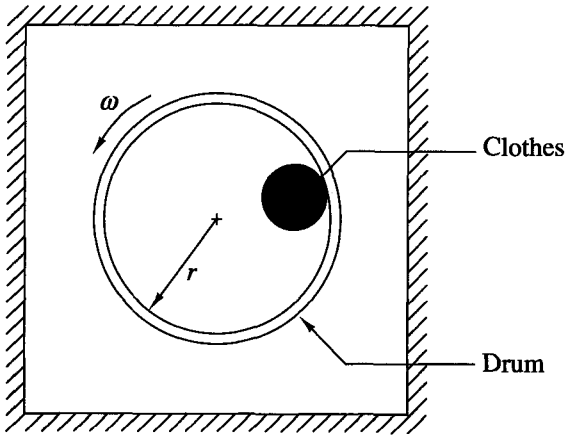


Figure P2.72

2.73. Consider the system of Problem 2.72 but refine the approximation as in Figure P2.73. The drum rotates within a nonrotating sleeve and the total assembly has a mass equal to 4 kg. The outer assembly is held in place by a spring, as shown in Figure P2.73. Note that motion is allowed only in the x direction. The spring constant k is 500 N/m. The ball of clothes is still 7 kg, and $\omega = 35$ rad/s. Determine the force applied to the ground.

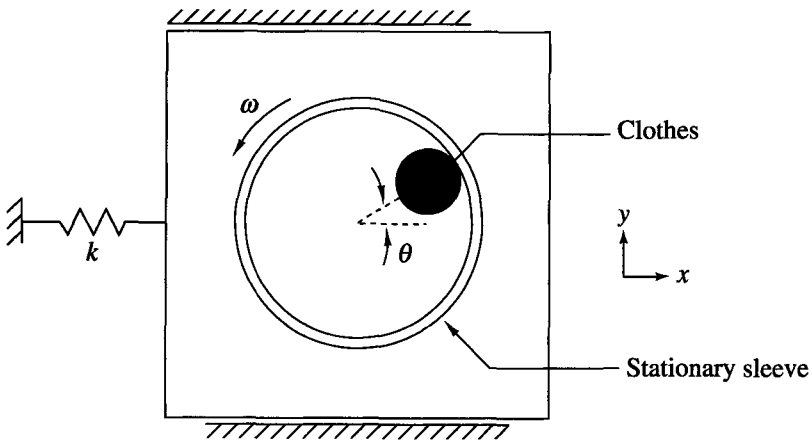


Figure P2.73

2.74. A simplified model of a rotating shaft within its bearings is shown in Figure P2.74. If the shaft is imbalanced, it will experience a time-varying force due to the rotating mass. The operating frequency is 60 rad/s, the support stiffness k_1 is 170,000 N/m, and the total rotating mass m is 100 kg. Determine the amplitude of vibration, given that at high frequencies ($\omega \to \infty$) the oscillation amplitude is equal to .001 m. You'll have to figure out l from the given data.

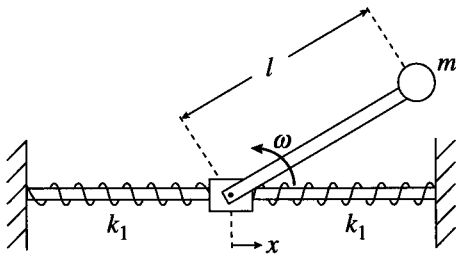


Figure P2.74

Section 2.10

2.75. All you have left from a vibration study is the scrap of paper illustrated in Figure P2.75. Can you use the data to reconstruct the peak response amplitude for the system? An answer that's within 5% is good enough. The system was a direct force excited, spring-mass system.

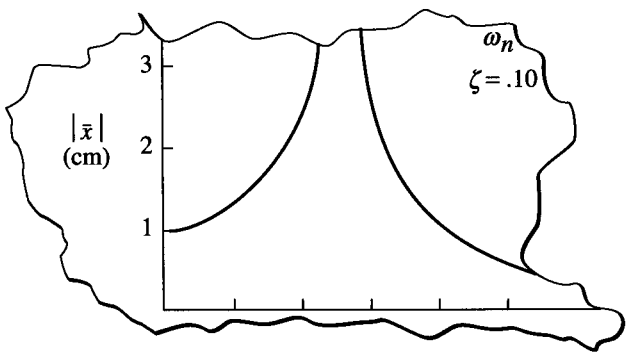


Figure P2.75

2.76. Assume that your car's shock absorber has different values of viscous damping on the extension stroke and on the compression stroke. If the value of the damping factor is $\zeta = .2$ during compression, what must it be on the extension stroke for the overall damping factor to be .4? (Ignore changes to ω_d due to varying damping rates; they'll be small for the values of ζ you're considering.)

2.77. Show that for small damping, the frequency at which the magnitude of the response of a seismically excited damped SDOF system (Figure P2.1) is maximized is the same as that of a direct force excited system.

2.78. Estimate the size of the peak amplitude response for

$$3\ddot{x} + 17.5\dot{x} + 4000x = 15\sin(\omega t)$$

without solving the equation exactly.

2.79. The peak amplitude response $|g(\omega_p)|$ is equal to 5 cm for a direct force excited (Figure P2.79), damped SDOF system. If you know that $m = .03$ kg, $c = .048$ N·s/m, and $k = 12$ N/m, can you determine $|g(0)|$?

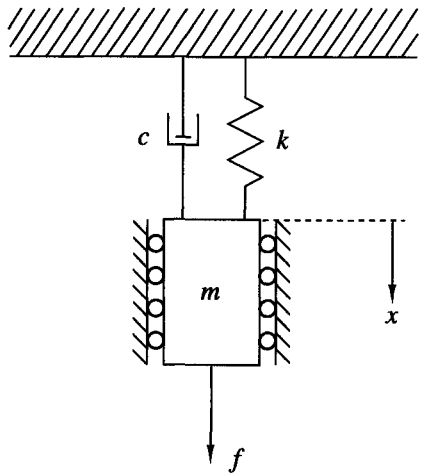


Figure P2.79

2.80. Can you deduce the natural frequency and damping factor of the system shown in Figure P2.80 from the following data? The damping factor is small ($\zeta \ll 1$) and the governing equation is

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2 x = \omega_n^2 \bar{f}\sin(\omega t) = f(t)$$

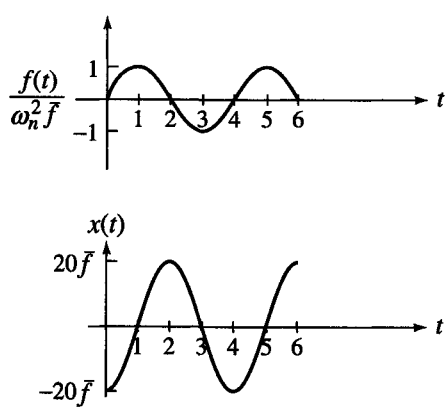


Figure P2.80

2.81. Ω_n is close to the average of Ω_{h_1} and Ω_{h_2} when ζ is small. Let's consider the case when ζ is relatively large. If $\zeta = .2$, what is the error in identifying Ω_n as equal to $\frac{\Omega_{h_1}+\Omega_{h_2}}{2}$?

2.82. Can you determine the value of ζ for the system and responses illustrated in Figure P2.82? The responses shown are two samples from a large number of tests. The governing equation is

$$m\ddot{x} + c\dot{x} + kx = f(t) = \bar{f}\sin(\omega t)$$

The test that produced the maximal amplitude response is the second one illustrated.

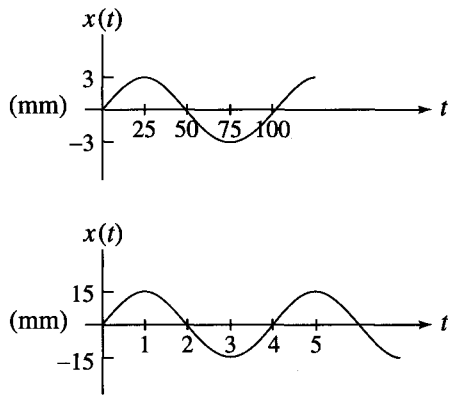


Figure P2.82

2.83. What is the error in using the half-power approximation $\zeta \approx \frac{\delta_h}{2\omega_n}$ if the actual damping factor is .3?

2.84. Why can't you use the half-power approach if $\zeta = .5$? Will the peak response approach, namely, $\frac{1}{2\zeta} = \frac{|g(\omega_p)|}{|g(0)|}$, work? How well?

2.85. Why will neither the peak response approach nor the half-power method work to determine ζ if the system's actual ζ is .8?

2.86. My dad bought a new car a few years back. Unfortunately, a vibration was noticeable when traveling at around 52 mph. The technicians took the car out for a test and gathered acceleration data from an accelerometer. Back at the shop they looked at the data and saw that a strong peak in the vibrational energy was present at 12 Hz when the car had been traveling at 52 mph, just as my father had claimed. I suggested to my dad that the vibration must be due to a resonance in the car that was excited because of the rotation of the tires. Why did I think this? (*Note*: The distance from the center of the wheel to ground for my dad's car was 1 ft.)

2.87. It's not necessary to have an entire cycle of damped response to determine the linear damping through the log decrement approach. Show how you can modify the approach detailed in Section 2.10 to produce an estimate for ζ based on the response information illustrated in Figure P2.87.

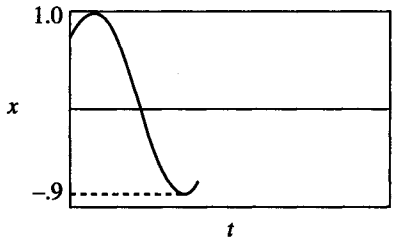


Figure P2.87

2.88. Was the system that produced the response illustrated in Figure P2.88 a linear one?

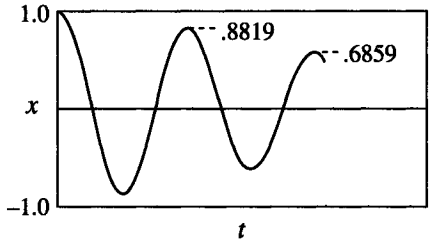


Figure P2.88

2.89. The dampers on automobiles are supposed to work in the same manner during compression and extension. You are examining one that owing to a manufacturing defect, is operative only during one half of the cycle (the compression stroke). If you know that the damping is viscous during compression and you also know that the amplitude drops from 1.0 to .15 over one cycle of free vibration, calculate the value of ζ that would be produced by a properly operating damper. The time elapsed between the two amplitude measurements was .5 second.

Section 2.11

2.90. In this exercise you'll determine the amount of energy dissipated over one cycle of forcing for Coulomb friction. Assume the response is given by $x(t) = a\sin(\omega t)$ and thus $\dot{x}(t) = a\omega\cos(\omega t)$. The equation of motion is

$$m\ddot{x} + \mu mg\,\mathrm{sign}(\dot{x}) + kx = f_s\sin(\omega t) + f_c\cos(\omega t)$$

The forcing has been broken into distinct sine and cosine components so that our assumption of a purely sinusoidal response for x will hold. Calculate the response and retain only those terms corresponding to sinusoidal oscillations at the frequency ω. This information, along with (2.11.8), will allow you to compute the energy dissipated over one cycle of motion.

2.91. Using the same approximation to the sign function as in Problem 2.90, show that the steady state amplitude of oscillation for a direct force excited mass with Coulomb friction is given by

$$a = \frac{\sqrt{\bar{f}^2 - \left(\frac{4\mu mg}{\pi}\right)^2}}{k - m\omega^2}$$

where $\bar{f}$ is the magnitude of the forcing and a is the response magnitude.

2.92. Using the response magnitude relation of Problem 2.91, determine what the magnitude of forcing must be for finite oscillations to exist. What if the forcing is below this limit?

2.93. Consider a direct force excited, spring-mass system with Coulomb friction; $k = 20,000$ N/m, $m = 6$ kg, $f = 100$ N, $\mu = .1$, $\omega = 12$ rad/s. Use the information in Problem 2.91 to determine the magnitude of the steady state response.

2.94. How much will the magnitude of oscillation be reduced if the system of Problem 2.93 has its coefficient of friction increased from .1 to .5?

2.95. How much will the magnitude of oscillation be altered if the system of Problem 2.93 has the forcing frequency increased to 30 rad/s?

2.96. Consider the static problem of a spring-mass system with Coulomb friction (Figure P2.96). If $k = 12,000$ N/m, $m = 10$ kg, and $\mu = .3$, what is the maximal initial deflection of the mass that will not result in subsequent motion of the mass?

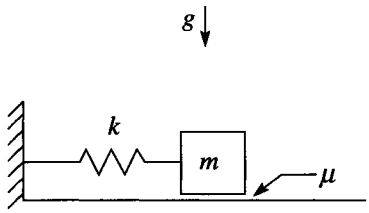


Figure P2.96

2.97. Consider a spring-mass system with Coulomb damping and direct force excitation. Let $f = 50$ N, $m = 1$ kg, $k = 1000$ N/m, and $\omega = 10$ rad/s. You've run two tests. You observe an oscillation magnitude of 55.5 mm for the first test and 55.6 mm for the second. Both readings seem to be within a reasonable experimental error. Calculate the value of μ that would correspond to these measurements and comment on whether determining μ from such a test is a reliable approach. Use the result from Problem 2.91.

2.98. Consider a spring-mass system with Coulomb damping and direct-force excitation. For $f = 25$ N, $\mu = .4$, and $m = 6$ kg, the system remains in static equilibrium. What should μ be reduced to if it is necessary to permit oscillations in the system? Use the result from Problem 2.91.

2.99. Consider a spring-mass system under direct force excitation with Coulomb damping. As in Problem 2.97, you've run two tests and recorded the oscillation magnitudes. For the first run you obtain a magnitude of 41.5 mm and for the second you obtain 42 mm. Use this information to calculate μ and comment

on whether, for these parameters, this is a reliable means of determining μ. $k = 1000$ N/m, $m = 2$ kg, $\omega = 23$ rad/s, $f = 5.1$ N.

Section 2.12

2.100. Consider the accelerometer described in Example 2.19. Assume that we've included a damper in the accelerometer, with $\zeta = .2$. With all other parameters the same, what will the percentage error be for the accelerometer's prediction of the magnitude of the acceleration? What phase error will this damping induce?

2.101. Consider the accelerometer described in Example 2.19. Assume that a viscous damper ($\zeta = .15$) is included in parallel with the spring. At what frequency will the phase difference between the actual acceleration and the accelerometer's measurement differ by 15 degrees?

2.102. If an accelerometer is being approximated as an undamped spring-mass system, what percentage of ω_n is the maximum frequency for which the sensor error is less than or equal to .5 percent?

2.103. Consider a seismometer for which $m = .25$ kg and $k = 250$ N/m. What is the lowest frequency of vibration at which the seismometer can be used if we require accuracy of at least 15%? Neglect any damping for this problem.

2.104. Consider Problem 2.103. If the mass of the seismometer is doubled, how much lower will the lowest allowable frequency be? Again neglect damping and allow a maximum of 15% error.